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Causal forest estimation of heterogeneous and time-varying environmental policy effects☆

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ABSTRACT

Empirically evaluating environmental policies requires grappling with impacts that exhibit not only cross-sectional heterogeneity, but also variation across time. Phased policy roll-outs offer opportunities for improvement across cohorts and policy effects can grow or decay, especially when natural processes are involved. Focusing on a subset of these factors can lead to erroneous inference, while considering them jointly magnifies specification challenges. To address these challenges, I extend and apply causal forests, a nonparametric method for estimating heterogeneous treatment effects, to simultaneously examine how effects vary across time. I first adapt causal forests to a panel setting with staggered policy introduction by incorporating dynamic selection assumptions and estimators. After illustrating the method's performance on simulated data, I use it to reanalyze how individual quota programs have affected fisheries catches around the world. Estimates reveal substantial heterogeneity and time dependencies and suggest that longer policy exposure may be less beneficial than previously thought. More generally, the approach has potential value for evaluating impacts of a range of environmental policies as well as environmental shocks.

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1. Introduction

Understanding who benefits from environmental policies and when those benefits accrue is essential to the evaluation of existing policies and the design and targeting of new ones. Gaining that understanding, however, is no simple task. Heterogeneity is pervasive: programs may induce a range of emissions abatement across firms, changes in ozone concentration across space, or modifications to energy consumption across individuals (Fowlie, 2010; Allcott, 2011; Auffhammer and Kellogg, 2011). Effects may also depend on duration of or time since an intervention, such as a new source of information on water or energy use (Ferraro et al., 2011; Allcott and Rogers, 2014; Martin and Rivers, 2018). Phased policy roll-outs may also result in different effects on those subjected to a policy later rather than earlier as policy implementation evolves (e.g., due to learning). Complicating matters, these factors can also interact: the time profile of policy effects may depend on cross-sectional characteristics like the growth rates of resources or local pollutant dynamics.

Studying policy effects that could vary in so many ways presents empirical challenges. First, identifying even average effects typically entails removing much of the variation in the data, through conditioning on observables, inclusion of fixed effects, use of only instrument-induced variation, or focusing on variation near a discontinuity. With potentially little residual variation

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remaining, the ability to answer more demanding questions about heterogeneity in policy effects may be limited by statistical power. Second, commonly used methods require pre-specifying how policy effects vary, either by interacting a policy indicator with specific variables in a parametric model, or by partitioning the data into groups and estimating a single average effect (parametrically or not) separately for each subsample. Both routes require important assumptions about the way policy effects vary, codified by choices of interaction terms or by selection of subsamples. Poor choices can lead to erroneous inference, including mistaking heterogeneity for dynamics or dynamics for learning. Estimating several models can help mitigate these issues, but doing so raises concerns about multiple testing and the universe of plausible models is often quite large. This is especially true in high-dimensional settings, where a large number of covariates, as well as policy duration and year of implementation, could modify the effect of the policy.

To address these challenges, I extend and apply a method based on random forests to jointly and nonparametrically examine heterogeneity, dynamics, and evolution of environmental policy effects. The proposed approach combines causal trees and forests (Athey and Imbens, 2016; Wager and Athey, 2018), which are designed for the cross-sectional study of heterogeneous treatment effects, with dynamic treatment frameworks (Lechner, 2002; Lechner and Miquel, 2010). The resulting method offers a unified way to estimate policy effects that vary cross-sectionally, with policy duration, and with the year a policy was introduced. Considering these dimensions jointly reduces the chances of mistaking one type of variation in policy effectiveness for another. Further, in contrast to the standard approaches mentioned above, researchers need not pre-specify the exact way in which the policy effect varies. Instead, causal forests partition the data adaptively based on variation in the treatment effect, effectively using degrees of freedom where they are most informative to aid in detection of heterogeneity and dynamics.

After outlining the approach and evaluating its performance via Monte Carlo exercises, I demonstrate its use by revisiting whether and when a common form of rights-based management – an individual quota (IQ) program – helps avoid resource collapse in fisheries around the world. Costello et al. (2008) examined this question on a global scale using standard panel and propensity score methods, finding that individual quota programs helped reduce the probability of collapse. That study also documented a linear trend in policy effectiveness: IQs in place for longer were more beneficial. Because the forest-based method I present leverages a similar identification strategy, using that study as a baseline allows me to highlight additional insights the new approach can offer.

Applying causal forests to this question paints a rich picture of heterogeneity and evolution of policy effectiveness through time. The estimated effect of IQ management on the probability of collapse among IQ fisheries ranges from a reduction of 11% to an increase of 7%, with an overall mean of 4.5% reduction.[1] The most beneficial policy effects occur in earlier implementations, for faster growing species, and in fisheries with higher, less stable catch levels just prior to the policy change. Many of these patterns are consistent with both population dynamics and optimal harvest policy. Previously depleted fisheries with low or no harvest may be optimally closed when IQ management begins (Reed, 1979), yielding apparent continuation of collapse. In addition, fisheries with high, stable catches prior to IQ adoption may already have managed harvests effectively (e.g. via fleet-wide quotas), such that the gains from IQ management are primarily economic rather than biological.

Collectively, the causal forest estimates also imply that the relationship between policy duration and impact is weaker than previously thought (Costello et al., 2008). In short, duration is correlated with policy start year as well as cross-sectional dimensions, and a model interacting treatment with duration alone may conflate dynamics with heterogeneity or changing implementations. To be clear, the intuition that duration may be confounded as a moderator of policy effects does not require causal forests and can be illustrated with simple parametric models (see Section 4.5.1). However, causal forests allow for flexible detection of variation in policy effects, so that the limited role of duration they uncover in this context strengthens insights from parametric models (see the online-only appendix, section S3 for an illustration).

Together, the approach and application offer three primary contributions. The first is methodological: I extend causal forests (Wager and Athey, 2018) to examine not only heterogeneity in policy effects, but also how those effects depend upon duration and time of first exposure. The modification leverages inverse probability of treatment weighting based on a dynamic treatment framework and associated conditional independence assumption (Robins et al., 2000; Lechner, 2002; Lechner and Miquel, 2010). The resulting method offers a unified approach to estimation of treatment effects that vary cross-sectionally, with duration, and with time of first exposure. Variation in effects across the time dimensions is, of course, quite different than cross-sectional heterogeneity, reflecting differences across exposures rather than varying responses to a single exposure. However, the unified approach allows for straightforward investigation of all three dimensions and their interactions.[2] Prior applied work has used causal forests only to examine cross-sectional heterogeneity, relying on pre-selection of different intervals to examine the role

[1] Some increases in collapse under IQs may also be artifacts of the catch-based collapse metric carried over from prior work. See Sections 4.2 and 4.7 for a discussion and examination of an alternate outcome.

[2] This approach does not identify structural parameters governing how policy effects vary with duration or start year; instead, it recovers two types of trend. The same is true of other reduced-form work examining trends in impacts of environmental policies (e.g., event studies like Grooms, 2015) or shocks (e.g., declining effects in Barreca et al., 2016). However, causal forests are likely to be particularly valuable precisely when structural estimation is impeded by a lack of theoretical guidance. The two are complementary: differences in policy effects across time can inform future work focused on mechanisms (again see Barreca et al., 2016).

of duration (Bertrand et al., 2017).[3]

The resulting method complements a literature examining nonparametric estimation of treatment effect heterogeneity. Recent examples leverage modified support vector machines (Imai and Ratkovic, 2013), least absolute selection and shrinkage operator (LASSO) estimators (Tian et al., 2014), Bayesian approaches (Taddy et al., 2016), and kernel regression (Abrevaya et al., 2015). While promising, those studies focus on heterogeneity alone and do not consider dynamics. Simpler approaches, such as on estimating conditional means of counterfactual outcomes via kernel regression, also have value. With appropriate covariate-specific and locally-adaptive bandwidths (e.g Racine and Li, 2004; Li and Racine, 2004; Hall et al., 2007), that approach would base estimates on local neighborhoods in a manner similar to regression trees. Importantly, the tree-based methods presented here adapt to variation in the estimated policy effect rather than the shape of the counterfactual functions themselves. Since the treatment effect may vary when one counterfactual does not, and the treatment effect may be constant when both counterfactuals are highly variable, the adaptivity of causal forests appears attractive.

As a second contribution, the estimated fisheries policy effects add to a large literature examining the effectiveness of individual quota programs. Of most direct relevance, many researchers have posited heterogeneity or studied it using conventional parametric methods, associating effectiveness with robustness of quotas (Mora et al., 2009), biological characteristics of the targeted species (Pinkerton and Edwards, 2009), country development (Ban et al., 2009), and the health of the fishery prior to adoption of individual quotas (Costello and Grainger, 2018).[4] Costello et al. (2008) also examine the role of policy duration but allow only linear trends; as explained above, those trends may partly reflect other forms of heterogeneity and changing effectiveness. A recent contribution from Isaksen and Richter (2019) confirms the general findings of Costello et al. (2008), and investigates several forms of heterogeneity using pre-specified sample splits, examining one dimension at a time. Non-causal regression trees and random forests have been used to identify fishery characteristics which predict favorable outcomes (Gutiérrez et al., 2011; Melnychuk et al., 2016), but not to estimate policy effects in a causal framework.[5] The results here complement those studies, highlighting potential nonlinearities and interactions in individual quota program effectiveness through use of causal forests. In particular, the causal forest approach strengthens evidence of confounding among policy effect modifiers suggested by both intuition and parametric models.

Beyond any insights specific to individual quota programs, the paper's third contribution is illustration of the application of (modified) causal forests to a question in the environmental domain. Causal forests have been adopted somewhat more rapidly in other fields, e.g., to study cross-sectional heterogeneity in effectiveness of summer jobs programs (Davis and Heller, 2017), wearable health gadgets (Handel and Kolstad, 2017), and public works employment (Bertrand et al., 2017). In the environmental realm, Prest (2017) examines individual electricity use in response to time-of-use pricing. However, none of these studies uses the proposed approach to jointly examine how policy effects depend on cross-sectional characteristics, duration, and start time.

The illustrative fisheries example highlights many features common across a range of empirical environmental work, and hence the potentially broad applicability of the approach. First, many environmental policies take effect in a staggered fashion, whether because different jurisdictions autonomously enact or modify the stringency of policies (e.g., Gallagher and Muehlegger, 2011), a federal government gradually cedes enforcement responsibility to more local levels (e.g., Grooms, 2015), or environmental conditions surpass key policy thresholds at different times across space (e.g., Auffhammer et al., 2009). Second, because environmental systems often exhibit natural dynamics due to decay or resource growth, the ability to examine time-varying policy effects is paramount. Many of those dynamics are nonlinear, so that even in low-dimensional settings like the fisheries application I examine, the flexibility afforded by causal forests should prove valuable.[6] Further, data rich, micro-level studies can even better leverage the value of causal forests in higher dimensions than is illustrated in the fisheries application here. Finally, while I focus on environmental policy analysis, the proposed method is equally applicable for quantifying effects of environmental shocks that exhibit both heterogeneity and dynamics, as in White (2017).

The remainder of this paper is organized as follows. In the next section, I briefly review causal trees and forests, then describe how they can be extended to examine variation across time. I then validate the resulting approach on simulated data before applying it to study the effectiveness of individual quota programs in fisheries. The final section concludes.

[3] Other applications of machine learning tools (e.g. LASSO) to examine treatment effect heterogeneity also use conventional sample splitting to examine differences in effects across time, e.g. Knaus et al. (2017).

[4] While these studies examine heterogeneity, none provide a preferable dataset for the exercise conducted here. The data in Mora et al. (2009) are cross-sectional, Pinkerton and Edwards (2009) and Ban et al. (2009) offer plausible hypotheses regarding heterogeneity but do not provide data to test them. Costello and Grainger (2018) do use panel data, but for a much smaller set of 375 fisheries – those in the RAM legacy database. Given my interest in joint heterogeneity, dynamics, and evolution of policy effects, a larger sample of treated fisheries is crucial.

[5] In addition, Gutiérrez et al. (2011) focus on co-managed rather than individual quota fisheries. Melnychuk et al. (2016) use only a cross-sectional dataset and conduct a predictive exercise only using data on fisheries managed via individual quotas (no 'untreated' observations).

[6] While causal forests are useful in high-dimensional settings, their handling of nonlinearity is both related and valuable. Let policy effects vary across eight dimensions as in the fisheries application (six fishery characteristics–five continuous and one binary–as well as policy start year and duration). To allow for flexibility in functional form, a regression approach might use dummies or polynomials in each continuous dimension, together with interactions. This nonlinear (basis) expansion of covariates on the right side of a regression affords flexibility by fitting a linear relationship in a higher dimensional space. Even with four dummies (e.g. quartiles) for start year and each of the five continuous fishery characteristics, there are already $2^*4^6 = 8,192$ different potential duration profiles of the policy effect function. The benefits of causal forests for higher-dimensional problems may thus be equally useful in lower-dimensional settings in which researchers do not have theoretical guidance on or wish to avoid functional form assumptions.

2. Quantifying heterogeneity, dynamics, and evolution of treatment effects

One way to understand how a policy's effects vary is to estimate its average impact on different subsamples. With an appropriate set of assumptions and a corresponding estimator, the resulting set of estimates can be interpreted as a step function approximation to the true policy effect. A key challenge, of course, is how to select a set of subsamples (locations of steps) so that the approximation of the policy effect function is "good" in some useful sense. Causal trees (Athey and Imbens, 2016) and causal forests (Wager and Athey, 2018) use a modified regression tree algorithm to automate selection of subsamples, favoring partitions in which treatment effects vary across subsamples (see equation (1) below). That approach is likely to be particularly useful in observational panel settings, in which substantial variation in the outcome and treatment variable are often removed to mitigate confounding. With limited residual variation, the detection of treatment effects and differences among them should be aided by the explicit focus of these methods on splitting the sample where doing so is most informative.[7]

Below I summarize these methods, then adapt them to a panel setting in which policy effects may also vary with policy duration and time of first exposure.

2.1. Background: causal trees and forests

Causal trees partition a dataset into subsamples, estimating a separate treatment effect per subsample, in order to minimize expected mean squared error (EMSE) between estimated and true treatment effects. Given a vector of covariates x and an unbiased estimator $\widehat{\tau}(x)$ of a conditional average treatment effect $\tau(x)$, EMSE is minimized by maximizing the quantity $Q \equiv E[\tau^2(x)] - E[Var(\widehat{\tau}(x))]$. A key contribution of Athey and Imbens (2016) is showing that Q can be estimated using properties of $\widehat{\tau}(x)$ alone:

$$\widehat{Q} \equiv \frac{1}{N_{tr}} \sum_i \widehat{\tau}^2(x_i) - \left(\frac{1}{N_{tr}} + \frac{1}{N_{est}} \right) \sum_i \widehat{Var}(\widehat{\tau}(x_i)), \tag{1}$$

where i indexes observations and N_{tr} and N_{est} are the number of observations in training and estimation datasets, respectively. The training dataset is used to identify the partition that maximizes $\widehat{Q}$, while the estimation dataset is used to estimate treatment effects taking that partition as given. This split-sample approach is termed "honest estimation" (Athey and Imbens, 2016).

There are two components to this criterion. The first term rewards identification of treatment effects that vary across subpopulations. The second term penalizes estimator variance within a subpopulation, which helps control too aggressively seeking heterogeneity through overfitting. Intuitively, the criterion is largest when the treatment effect is well approximated by a step function: constant within but varying across subpopulations.

While $\widehat{Q}$ thus defines a sensible ranking of potential partitions, the $\widehat{Q}$-maximizing partition must still be identified. For that task, the classification and regression tree (CART) algorithm can be applied with splitting and cross-validation based on variants of $\widehat{Q}$ instead of in-sample prediction error metrics. In short, the algorithm begins with a single estimate and $\widehat{Q}$ for the entire sample, then recursively splits the training sample in two based on whether the value of a single covariate for an observation lies above or below a threshold. The algorithm exhaustively searches through all such sample splits at each stage to find that which increases $\widehat{Q}$ the most. For more details, see Athey and Imbens (2016) or the large literature on conventional regression trees (e.g. Fisher, 1958; Breiman et al., 1984; Chou, 1991).

Wager and Athey (2018) extend this idea by building many such step-function approximations to the treatment effect function: a forest of trees. While a single step-function approximation (tree) is highly variable, averaging over many different approximations can provide more stable results. The random forest algorithm (Breiman, 2001) does exactly that, building each tree from a dataset resampled from the original data, considering a random subset of variables each time the data is split, and averaging over tree estimates to generate a single forest estimate. Wager and Athey (2018) adapt the random forest procedure to use causal trees and establish key asymptotic properties of the resulting causal forest. Because forests limit overfitting via averaging, the second term in (1), which serves a similar purpose, is omitted when considering partitions.

2.2. Estimating policy effects which vary both cross-sectionally and temporally

Causal trees and forests can be readily adapted to study policy effects that may vary not only with cross-sectional characteristics, but also with both policy duration and time of first policy exposure. The proposed method retains the broad structure of a causal forest: a set of causal trees is constructed from resampled datasets, and the forest estimate is the average of relevant estimates across the trees. Within each tree, the data are partitioned into subsamples, a single policy effect is estimated for each subsample using an existing econometric estimator.

Four main modifications are needed to accomplish this goal. First, the maintained identifying assumptions should reflect a policy adoption (selection) process that occurs over and may vary with time. Second, the estimator used in each subsample must

[7] Intuition for this advantage can be gleaned from minimum detectable effect (MDE) calculations in a panel context, e.g., Burlig et al. (2019). The MDE is inversely related to variation in the treatment variable; removing variation by adding controls or fixed effects (or choosing smaller subsamples) increases the MDE, making detection more challenging.

be updated to reflect the dynamic nature of both treatment and the corresponding effects. Third, the partitioning procedure must be able to split data (define subsamples) based on policy duration or time of first exposure. Finally, resampling used to generate data for each tree should reflect the panel nature of the data, and inference procedures should be updated accordingly. These modifications are all relatively straightforward, and I address them below in turn after setting up the estimation objective.

2.2.1. Estimation target

The goal of estimation in this paper is to identify the average effect of a policy on an outcome of interest, allowing the effect to vary with pre-policy characteristics of an affected entity, the time the policy first went into effect ("start time"), and how long an entity has been subject to the policy ("duration"). For this purpose I leverage existing dynamic treatment frameworks (Robins et al., 2000; Lechner, 2002; Lechner and Miquel, 2010). Those frameworks are quite general, allowing comparison of expected potential outcomes corresponding to two arbitrary treatment sequences, each of which may contain multiple (potentially non-adjacent) treated periods. Here, I restrict the analysis in two ways to match common environmental policy settings. First, I consider only policies that remain in effect once an entity is first subjected to them, which holds in the empirical application later. Second, I focus on comparisons between treatment of some duration and a counterfactual of no treatment, which is the target of most existing environmental policy analyses.

With this background the estimation target can be defined. I focus on the conditional average treatment effect (CATE), with conditioning occurring on pre-treatment characteristics alone. Suppose we observe outcomes Y_{it}, characteristics X_{it}, and a binary treatment status T_{it} (= 1 if treated) for entity i at time t. By the assumption that treatment is permanent, $T_{it'} \geq T_{it}$ for $t' > t$. Denote the history of any variable by its boldface version, e.g., $\mathbf{X}_{it} = \{X_{i1}, \dots, X_{it}\}$ and the duration (number of periods) of policy exposure by $D_{it} = \sum_{t'=1}^{t} T_{it'}$. Let the potential outcome (Rubin, 1974) for unit i at time t exposed to d periods of treatment as of t be $Y_{it}(d)$. Denoting the potential start time of treatment by $s = t - d + 1$, the conditional average treatment effect (CATE) is:

$$\tau^*(x, d, s) \equiv E[Y_{it}(d) - Y_{it}(0)|\mathbf{X}_{is-1} = x, s = t - d + 1]. \tag{2}$$

In sum, the quantity of interest is the expected difference in outcomes in period t after d years of policy exposure compared to a counterfactual of no exposure, conditional on pre-treatment characteristics. I express the CATE in terms of s and d rather than t and d to facilitate more natural discussion of how effects vary by cohort.

In practice, sufficient data may not exist to credibly estimate (2) for all unique combinations of x, d, and s. In such cases, an alternate estimation target is to compute an average effect over a broader subpopulation $X_{is-1} \in \mathcal{X}$ and a broader set of exposures defined by starting times $s \in \mathcal{S}$ and durations $d \in \mathcal{D}$:

$$\tau(\mathcal{X}, \mathcal{D}, \mathcal{S}) \equiv \frac{1}{|\mathcal{D} \times \mathcal{S}|} \sum_{d \in \mathcal{D}} \sum_{s \in \mathcal{S}} E[Y_{it}(d) - Y_{it}(0)|\mathbf{X}_{is-1} \in \mathcal{X}, s = t - d + 1]. \tag{3}$$

Note that averaging over $\mathbf{X}_{is-1} \in \mathcal{X}$ depends on the distribution of characteristics in the subpopulation $\mathcal{X}$, but averaging over s and d weights each potential exposure equally. Further, the characteristics $\mathbf{X}_{is-1}$ used to define the subpopulation of interest are restricted to those observed prior to the start of the potential treatment. In contrast, selection into treatment is dynamic and allowed to depend on characteristics occurring after period $s - 1$. I turn next to these issues of selection and identification.

2.2.2. Identifying assumptions

Since it is impossible to observe i in period t under two different treatment histories, further assumptions are needed to estimate (3). First, I adopt the Weak Dynamic Conditional Independence Assumption (w-DCIA) (Robins et al., 2000; Lechner and Miquel, 2010), which states that policy exposure in a period is as good as random after conditioning on sufficient information.[8] The conditioning information includes histories of both policy exposure and covariates; the latter may include past outcomes and variables influenced by past outcomes. Formally, the w-DCIA states:

$$Y_{it'}(d) \perp\!\!\!\perp T_{it}|\mathbf{T}_{it-1}, \mathbf{X}_{it-1};\ \forall t' > t, d \geq 0 \tag{4}$$

The second assumption pertains to overlap, stating that there must be a positive probability of observing entities exposed to the different policy durations needed to identify the effect(s) of interest. Because I focus on policies that, once in place, remain in place, there is zero chance of observing $T_{it} = 0$ if $T_{it-1} = 1$. However, that constraint in no way precludes estimation of (3) [9]; we only require the following restricted overlap assumption:

$$P(T_{it} = 1|\mathbf{T}_{it-1} = \mathbf{0}, \mathbf{X}_{it-1} = \mathbf{x}) \in (0, 1). \tag{5}$$

In short, for entities not yet subject to the policy, the entity must neither remain untreated nor begin treatment with certainty.

On a final note, effects $\tau^*(x, d, s)$ for which combinations of d and s are not directly observed in the dataset are unidentified without further assumptions. For example, outcomes for $s = 2018$ and $d = 5$ have not yet occurred. While the estimation approach that follows will produce estimates for such hypothetical treatments via constant extrapolation, I do not report

[8] If (3) instead defined the subpopulation of interest using covariates in period s or later, a strong(er) version of the dynamic conditional independence assumption would be required, essentially ruling out impacts of treatment on variables used to define the subpopulation of interest.

[9] It is impossible, however, to estimate the effect of, e.g. four continuous years of treatment as compared to three years of treatment followed by one untreated year.

them due to lack of identification. The same issue precludes some exercises in comparative statics for $\tau^*(x, d, s)$ for the earliest observed start year and longest duration, denoted $\tau^*(x, d_{max}, s_{min})$. In that case, $\tau^*(x, d_{max}, s)$ is unidentified for all s other than s_{min}: larger s means we have not yet observed outcomes for d_{max}, while smaller s, by definition, does not exist in the dataset. As a result, the effects of a marginal change in s on the policy effect at $s = s_{min}$ and $d = d_{max}$ are unidentified without further assumptions.

2.2.3. Estimation

Under assumptions (4) and (5), the effect of interest (3) can be estimated for any $(\boldsymbol{\mathcal{X}}, \boldsymbol{\mathcal{D}}, \boldsymbol{\mathcal{S}})$ tuple in many ways; I use inverse probability of treatment weighting (IPTW). The intuition behind IPTW is analogous to that for weighting estimators in static settings: the estimator 'undoes' the reweighting introduced by the selection process in order to better reflect the underlying population. In a dynamic setting, the relevant weights correspond to the inverse probability that an entity experienced its entire sequence of policy exposure. Those overall probabilities are the product of conditional probabilities of observed treatment in each period, and, under the maintained assumption of permanent treatment, can be written as follows:

$$p_i(d, s, t) = \begin{cases} P(T_{is} = 1 | \mathbf{T}_{is-1} = \mathbf{0}, \mathbf{X}_{is-1}) p_{i0}(s-1) & \text{if } d > 0 \\ p_{i0}(t) & \text{if } d = 0 \end{cases} \tag{6}$$

where

$$p_{i0}(l) = \prod_{t'=1}^{l} P(T_{it'} = 0 | \mathbf{T}_{it'-1} = \mathbf{0}, \mathbf{X}_{it'-1}). \tag{7}$$

Intuitively, the probability of nonzero policy duration is the probability of beginning treatment in the specified start time and not before, while the probability of zero duration is simply the probability of not being treated by time t. Each conditional probability can be estimated and an overall estimate $\widehat{p}_i(d, s, t)$ can be constructed accordingly.

Both overlap and balance assessments should be conducted using these probabilities before constructing weights. In keeping with overlap assumption (5), overlap can be assessed using standard methods in terms of the conditional probability of transitioning into treatment during a period. As in cross-sectional analyses, it may be prudent to use trimming to cap the influence of observations with extremely low estimated $p_i(d, s, t)$.

By analogy to the estimand (3), the IPTW estimator in a subsample defined by a combination of $\boldsymbol{\mathcal{X}}, \boldsymbol{\mathcal{D}}$, and $\boldsymbol{\mathcal{S}}$ can now be defined:

$$\widehat{\tau}(\boldsymbol{\mathcal{X}}, \boldsymbol{\mathcal{D}}, \boldsymbol{\mathcal{S}}) = \frac{1}{|\boldsymbol{\mathcal{D}} \times \boldsymbol{\mathcal{S}}|} \sum_{d \in \boldsymbol{\mathcal{D}}} \sum_{s \in \boldsymbol{\mathcal{S}}} \widehat{\tau}(\boldsymbol{\mathcal{X}}, d, s). \tag{8}$$

Here $\widehat{\tau}(\boldsymbol{\mathcal{X}}, d, s)$ is a standard IPTW estimator for a single policy exposure (s, d).[10]

With the estimator defined, the causal forest splitting procedure can be applied to splits on s or d with one final modification. Because s and d represent *potential* start time and *potential* duration, they do not exist in observed data. Assigning d and s is simple for treated observations: set $d = D_{it}$ and $s = t - d + 1$. In contrast, since control observations have $D_{it} = 0$, it is not obvious which values of d and s should be assigned. One option, which I adopt, is to replicate control observations for any values of s and d for which the control observation could plausibly aid in counterfactual estimation. For example, and untreated observation in 2017 could act as a control for $d = 1$ and $s = 2017$, or as a control for $d = 2$ and $s = 2016$. This approach is similar to matching with replacement, where the same control can contribute to multiple counterfactual estimates. Mechanically, this is accomplished as a data pre-processing step before the tree procedure starts.[11]

2.2.4. Panel unit subsampling

In a panel setting, unobserved determinants of an outcome of interest are likely to be correlated across time. As a result, I use a panel subsampling approach in which all observations from a given panel unit are either included or excluded from the sample used to construct each tree. This approach differs from the observation-level subsampling used in Wager and Athey (2018) but is suitable for resampling panel data (Kapetanios, 2008).

2.2.5. Inference

For inference, I use the infinitesimal jacknknife approach from Wager and Athey (2018), adapted to account for the panel nature of the data and my resampling approach. Two straightforward changes are required. First, the covariances entering the infinitesimal jackknife relate individual tree estimates and the inclusion of entire cross-sectional units rather than individual observations. Second, I use the (smaller) number of panel units in a sample rather than the number of observations in finite

[10] While causal forests do not use the second term in (1), for a single causal tree, $var(\widehat{\tau})$ could be estimated via, e.g. Imbens (2015).

[11] To save on computation, each untreated observation could be replicated for a random subset of the (d, s) pairs for which it could serve as a potential control.

sample corrections. Doing so inflates estimated variances and confidence intervals to account for potential intertemporal correlation among observations from a panel unit.[12]

A related approach can also be used to conduct inference on differences in treatment effects between two groups $(\mathcal{X}, \mathcal{D}, \mathcal{S})$ and $(\mathcal{X}', \mathcal{D}', \mathcal{S}')$. Each tree provides an estimate of the difference in treatment effects between those groups, and so an infinitesimal jackknife approach can be applied to those estimated differences. Accounting for covariance in estimates across subpopulations in this way is likely to be especially important since each tree in a forest generates a single estimate for all elements in a given subsample.

2.2.6. Handling unobservables

A clear threat to identification under w-DCIA is the presence of confounding unobservable factors, such as unit-specific effects or time effects. Time-varying but cross-sectionally uniform unobservables are of less concern given that (8) first constructs period-specific estimates and then averages over those estimates in $\mathcal{D} \times \mathcal{S}$. A complementary approach makes use of double differencing as a data pre-processing step. Let φ_t and μ_i represent additive period and unit effects, respectively. Given their additive nature, a natural approach is to eliminate these effects via double differencing. Any differencing must account for the staggered nature of treatment: unit-level means depend on the unit-specific frequency of treatment, while period-level means depend on the period-specific incidence of treatment. To this end, I make use of never-treated units in the following way. First, I average outcomes within period t across never-treated units, and subtract those means to remove the effect of φ_t. Then, the pre-treatment, within-unit mean of that intermediate outcome (with time effects removed) can be subtracted out to remove unit-level effects without making the demeaning process dependent on the frequency of treatment for a unit. The resulting transformed outcome can be used as the outcome of interest in the algorithm described above.[13] An alternative would be to estimate a regression with time and unit dummies in each subsample. However, data pre-processing via double differencing retains the computational advantages of a simple comparison of weighted means in each element of a candidate partition.

3. Monte Carlo evidence of estimator performance

To motivate use of the estimator in the fisheries application, I first evaluate its performance on three artificial data generating processes. All share a policy effect that varies with duration, start time, one of two cross-sectional characteristics, and an interaction between that cross-sectional characteristic and duration. The second cross-sectional characteristic does not influence the policy effect. Specifically, the policy effect takes the following form:

$$\tau(x_1, x_2, d, s) = 5 \cdot \mathbb{1}(x_2 \leq 0.5) + d - 0.025d^2 + 0.001d \cdot \mathbb{1}(x_2 \leq 0.5) + 0.2(s - 1980). \tag{9}$$

This choice of policy effect allows examination of the method's ability to deal with several features: an irrelevant cross-sectional characteristic, a non-smooth influence of a cross-sectional characteristic on the policy effect, nonlinearities in the effect of duration, separate effects of start year and duration, and an interaction between cross-sectional characteristics and duration. In particular, one potential advantage of the forest-based approach is its ability to identify jumps in policy effectiveness without pre-specification of the location of those jumps, as a researcher imposing dummy variables would have to otherwise do.

I construct three types of dataset using this policy function. The first is unconfounded, while the second and third have a selection probability that increases with the covariate influencing treatment:

$$P(T_{it} = 1 | T_{it-1} = 0, x_1, x_2, t) = 0.02 + 0.01x_2. \tag{10}$$

The data are generated for 1000 cross-sectional units and 30 years using these functions, with x_1 and x_2 sampled uniformly from the unit interval and Gaussian noise ϵ_{it} with mean zero and unit variance added to the outcome:

$$y_{it} = T_{it}\tau(x_1, x_2, D_{it}, S_{it}) + \epsilon_{it} \tag{11}$$

Note that the covariates influence the outcome only through their impact on the policy effect. The third group of simulations differ in that ϵ_{it} is allowed to be correlated across time to approximate a common concern with panel data. Specifically, $cov(\epsilon_{it}, \epsilon_{is})$ is 0.3 for $|s - t| = 1$, is 0.1 for $|s - t| = 2$, and is 0 otherwise. In contrast, in the first two groups of simulations the ϵ_{it} are independent. The modified causal forests perform well for all three groups of simulations, capturing heterogeneity in the policy effect and achieving coverage at near-nominal rates. To support this claim, I generated 20 samples per data generating process, and for each sample, I estimated a forest with 1000 trees. Example estimates from one forest for the third group of simulations – with both nonrandom selection and within-unit error correlation – are depicted in Fig. 1; results for the first two cases are qualitatively similar. The estimated policy effects (black) broadly track the patterns in the true effects (red) across different covariate combinations, start years, and durations, also correctly ignoring the irrelevant regressor x_1. Forests perform best when both covariates and the policy start year are toward the middle of their observed ranges and underlying heterogeneity is less

[12] An alternative, computationally intensive method is to bootstrap the entire forest-building procedure and resample entire forests. See, e.g., Asher et al. (2016) and Knaus et al. (2017) for related approaches.

[13] Suppose $Y_{it}(d, s) = \mu_i + \varphi_t + \tau(d, s)$. Estimate $\widehat{\varphi}_t = \frac{1}{|i:T_{it}=0\ \forall t|}\sum_{i:T_{it}=0\ \forall t} Y_{it}$ and define $\widetilde{Y}_{it} = Y_{it} - \widehat{\varphi}_t$. Estimate $\widehat{\mu}_i = \frac{1}{|t:T_{it}=0\ \forall t|}\sum_{t:T_{it}=0\ \forall t} \widetilde{Y}_{it}$ and define $\ddot{Y}_{it} = \widetilde{Y}_{it} - \widehat{\mu}_i$. Use $\ddot{Y}_{it}$ as the outcome of interest.

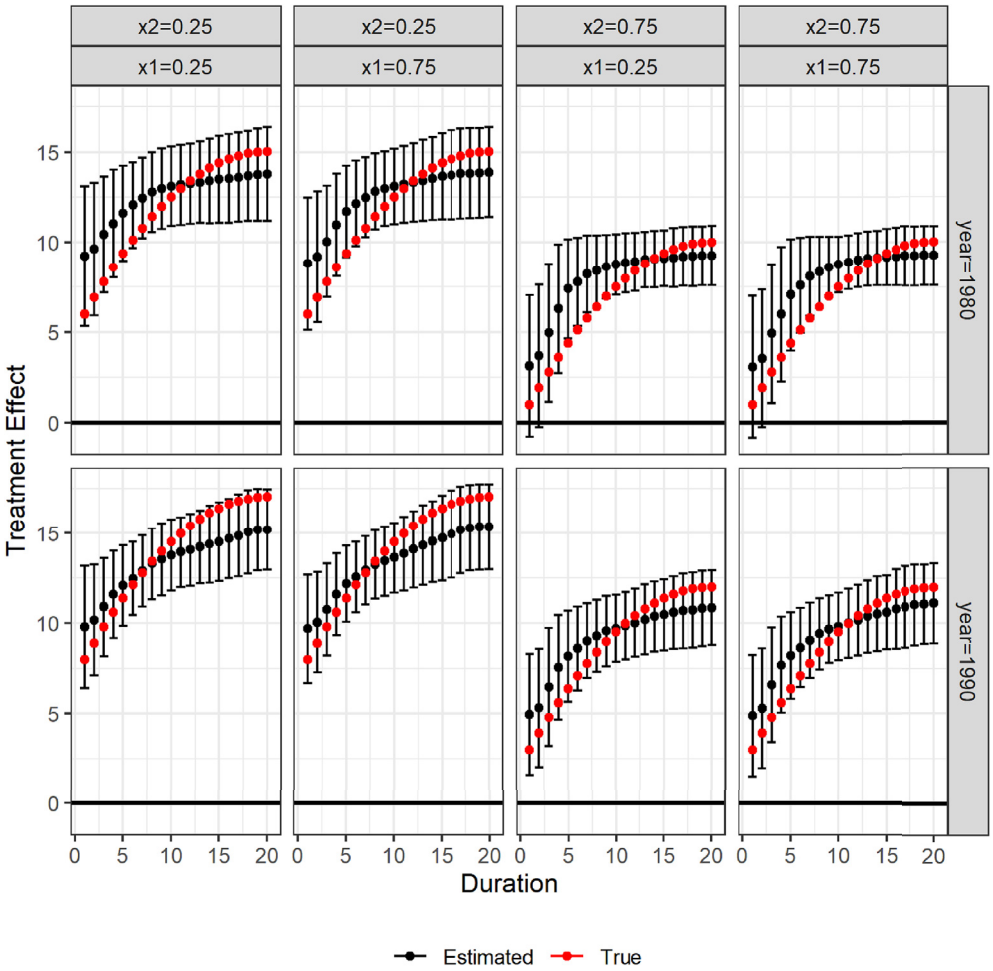

Fig. 1. Estimation results for nonrandom selection into treatment via (10) and the policy effect function (9). Each panel (combination of start year and covariates) depicts true (red) and estimated (black, with 95% confidence intervals) policy effects as a function of duration (x axis). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

pronounced (lower right panels). These patterns are to be expected: tree-based methods use piecewise-constant approximations, such that the estimated policy effects will be flatter than true effects, especially near the edges of the covariate and time space.

Confidence intervals constructed using the panel-adapted infinitesimal jackknife also provide near-nominal coverage. Across the 20 forests, coverage of the 95% confidence intervals at 500 randomly generated test points is 0.945 for the unconfounded data, 0.944 for nonrandom selection as in (10), and 0.949 for the case with both nonrandom selection and correlated errors. These near-nominal coverage rates, as well as the method's ability to separate out effects of duration from start year and cross-sectional characteristics, will be important when re-evaluating claims that the effectiveness of fisheries IQ programs improves with duration.

To more fully explore the ability of causal forests to detect heterogeneity and dynamics in policy effects, I conduct several supplementary simulations, with results presented in the online-only appendix (Section S1). Reassuringly, the causal forest approach is able to detect an interaction between a relevant regressor and duration across a range of interaction strengths and correlations with irrelevant covariates (Section S1.1). As with any approach, however, it has limitations: performance begins to degrade when heterogeneity occurs in a small part of the covariate space (such that few observations can contribute to estimation), especially near the edge of the covariate space. See Section S1.2 of the online-only appendix for details.

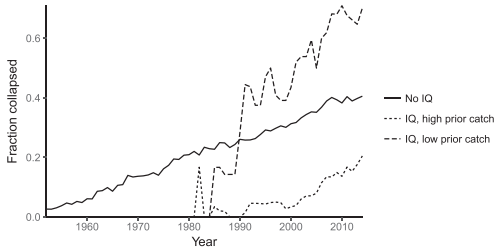

Fig. 2. Fraction of fisheries collapsed in a given year in three groups: those not under IQ management (solid), those under IQ management with low catch ($\leq$ 20% of historical max) just prior to IQ implementation (long dash), and those under IQ management with high catch ($>$ 20% of historical max) just prior to IQ implementation (short dash).

4. Application: individual quota markets in fisheries

4.1. Background

Since the mid-1970s, a number of commercial fisheries around the world have shifted to management based on individual quotas, which allocate rights to catch fish to specific entities (individuals or groups).[14] For example, in many fisheries a governing body initially allocates or auctions off shares of harvest quota, and shares are converted to pounds of fish each year when annual fishery-wide quotas are set. Rights-based management has the potential to improve both economic and biological outcomes by addressing dynamic externalities, reducing incentives to overinvest in resource extraction capital, and increasing the value of landings by lengthening harvest seasons (Grafton et al., 2000). However, both biological and economic effects of IQ adoption are likely to vary across fisheries and through time: optimal management and the rents that flow from it will depend upon initial resource conditions and the biology governing system dynamics, among other factors.

I use this empirical context as a test case for the modified causal forest approach described above, asking how adopting an individual quota policy impacts the probability of fishery collapse. This empirical setting is attractive for three main reasons. First, a high-profile benchmark study used conventional parametric propensity score and fixed effects methods to examine whether IQs reduce the probability of fishery collapse, with collapse defined as catch falling below 10% of its historical maximum (Costello et al., 2008). Results indicated that IQ adoption reduces probability of collapse on average, but the specifications allowed for neither heterogeneity nor variation in policy effects based on start date, and effects were allowed to depend on policy duration in a linear fashion only. Because the tree-based methods used here rely on similar identifying assumptions, a comparison of results should offer illustration of what new insights causal forests might provide.[15]

A second reason this empirical setting is useful is that prior work has indicated IQ effectiveness is heterogeneous, suggesting dependency on, e.g., social factors (Gutiérrez et al., 2011), scientific robustness of quotas set (Mora et al., 2009), biological characteristics of targeted species (Pinkerton and Edwards, 2009), and country development (Ban et al., 2009). More recent work suggests that quota stringency, and hence catch-based outcomes, depends upon the strength of property rights (Costello and Grainger, 2018).

Consistent with prior work, Fig. 2 presents suggestive evidence of heterogeneous effects. In particular, Fig. 2 depicts trends in collapse among fisheries not under IQ management, and two groups under IQ management: fisheries with low versus high catch just prior to IQ implementation. Differences between the IQ groups and the non-IQ group vary through time, and there are also clear differences in trends across the two IQ groups.[16] Those differences in collapse trends are consistent with heterogeneous effectiveness of IQ programs.

Finally, the effects of IQ management on the probability of collapse may plausibly vary with both the year a fishery transitioned to IQs and the duration of policy exposure. Because IQ management has been introduced in fisheries over the course of over four decades, there has been ample opportunity for learning and improvement in policy implementation. In addition, the definition of fishery collapse and the dynamics of biological populations suggest noticeable policy-induced changes in the probability of collapse may not appear immediately after a fishery transitions to IQ management. The likely evolution and dynamics of policy effectiveness in this context, together with heterogeneity, are exactly what the proposed method is designed to uncover.

The specific forms of heterogeneity and dynamics investigated here can be motivated using simple models and associated optimal harvest rules derived in earlier work (Reed, 1979; Costello and Polasky, 2008). Under suitable conditions, optimal harvest takes a constant escapement form, in which no harvest occurs if the population is below a target level; otherwise the stock is harvested until it reaches that level. The target escapement satisfies a golden rule in which the marginal increment to growth

[14] The analysis here does not distinguish between tradable and non-tradable forms of harvest rights, focusing broadly on any policy that assigns an individual right to a flow benefit.

[15] While valuable, this empirical setting does not fully exploit the potential benefits of causal forests. The relatively small set of covariates does not highlight potential gains from causal forests in detecting heterogeneity in high-dimensional settings.

[16] Some of the apparent increase in collapse across all groups may be an artifact of the collapse metric carried over from prior work. See Section 4.2 for a discussion of these concerns and their mitigation.

from leaving a fish in the water is equal to the rate of return from harvesting that fish and investing the proceeds:

$$f'(e^*) = \frac{1}{\delta}. \tag{12}$$

Here f is an increasing and concave differentiable growth function, e^* is optimal escapement (fish remaining after harvest), and δ is the discount rate. This rule holds if the number of fish available prior to harvest is above e^*; otherwise the fishery is closed.

Several insights relevant to the current analysis can be gleaned from that simple rule, provided that managers of IQ programs choose quotas in an approximately optimal manner. First, faster growing species (higher f') will yield larger fishable populations sooner, allowing for faster increases in harvest and reductions in the probability of 'collapse' (harvest below 10% of its historical maximum). Second, if the fishery begins at a depleted state (with populations well below e^*), it may be optimal to close the fishery for several years until the fishable population rises above e^*. We should thus expect policy effects to depend upon both the size of the fish population prior to harvest as well as the duration of IQ management. Third, nothing about (12) requires reaching e^* using individual quotas; a fishery with well-enforced fleet-wide quotas could be in equilibrium, harvesting a similar amount and leaving close to e^* each year. We might thus expect the biological gains (i.e. reductions in collapse) from IQs to be smaller for fisheries already having stable harvests near an equilibrium state. In contrast, if quotas under IQs are not well enforced due to poor governance, or because species are highly migratory and span multiple jurisdictions, we might expect any reductions in the probability of collapse to be more modest. Finally, the quality of and resources devoted to enforcement are also likely influenced by species value; I use species trophic level as a (positively correlated) proxy for that value.

4.2. Data

To examine heterogeneity, dynamics, and changing effectiveness of IQ programs, I assemble a global panel dataset of fisheries catch, IQ management status, and species characteristics ranging from 1950 to 2014. For consistency with Costello et al. (2008), a fishery is defined by a species and Large Marine Ecosystem (LME) pair. For the purposes of this analysis, each LME defines a spatial unit in which several fisheries operate. Catch records (tons per fishery per year) come from the Sea Around Us project, IQ policy adoption indicators are derived from a database complied by the Environmental Defense Fund, fish growth rate parameters (Von Bertalanffy, year^{-1}) and trophic level, where available, are collected from FishBase. Highly migratory status (yes/no) and a World Bank governance indicator area-weighted to the LME level are from Isaksen and Richter (2019). Where a species is subjected to multiple forms of management within a single LME, the earliest date of IQ implementation for a species in an LME is used to determine IQ status for an observation. Records are matched across data sources using combinations of species, location, and year. I construct the dataset to serve as an updated and expanded version of the data used in Costello et al. (2008). See the Appendix for additional details.

The data map neatly to the notation from earlier with i indexing fisheries and t years. The policy variable T_{it} is one if fishery i is subject to IQ management in year t, with D_{it} the cumulative number of years of IQ management for fishery i as of year t. The outcome of interest, Y_{it}, is a binary variable taking value one if and only if a fishery is considered collapsed, which occurs if the catch c_{it} for fishery i in year t is less than 10% of the maximum catch observed in fishery i in all years $t' < t$, i.e., $Y_{it} \equiv \mathbb{1}(c_{it} \leq 0.1 \cdot c_{it}^{max})$ with $c_{it}^{max} \equiv \max_{t':t'<t} c_{it'}$. The covariates X_{is} with which the IQ effectiveness is allowed to vary are one year lagged relative catch (c_{is-1}/c_{is-1}^{max}), one year lagged relative catch trend ($(c_{is-1} - c_{is-2})/c_{is-1}^{max}$), the Von-Bertalanffy growth rate, trophic level, and highly migratory status of the species, and the governance indicator. Because estimates of the underlying population are not available for all fisheries in the dataset, the lagged catch variables are used as proxies for that biological information. For example, a fishery with low and unchanging catch prior to IQ management is likely to be in a depleted state, either because that is all the population can support or the existing management regime has intentionally restricted catch.

The collapse metric just described has well-known limitations. Because it relies only on catch data, it may not align well with other measures of collapse that incorporate population estimates or other management data (de Mutsert et al., 2008; Branch et al., 2011). Further, because historical maximum catch cannot decline through time, rates of collapse using the approach outlined above will tend to increase over time due to random sampling even if the underlying population is relatively stable (Wilberg and Miller, 2007).

I offer four responses to these concerns. First, despite the metric's limitations, using it allows for clean comparison with prior work to illustrate the potential benefits of the proposed causal forest approach. Second, catch data are available for far more fisheries than independent biomass estimates, and a larger sample is essential to examining heterogeneity. Third, some of the mathematical concerns with the metric are mitigated by the estimator used here, which compares IQ and non-IQ fisheries in the same year (see (13)), making common trends in collapse metrics less concerning. Finally, I also present results using catch as an outcome (Section 4.7) without any reference to collapse. Catch is an important outcome of interest to stakeholders, regardless of its utility as an indicator of collapse.

Prior to estimation, the dataset is cleaned and filtered in two ways to make it comparable with that used in Costello et al. (2008). First, attention is restricted to fisheries occurring in LMEs which contain at least one fishery under IQ management during the study period in order to increase similarity between IQ fisheries and control fisheries. Second, the data are filtered to complete records containing all outcomes and explanatory variables.

Table 1
Summary statistics.

	All fisheries		IQ fisheries	
	mean	sd	mean	sd
Collapsed	0.23	0.42	0.13	0.33
Collapsed Under IQ			0.18	0.38
Relative Catch	0.50	0.37	0.58	0.34
Relative Catch Under IQ			0.45	0.32
Pre-IQ Relative Catch			0.54	0.35
Pre-IQ Relative Catch Trend			−0.02	0.17
Von-Bertalanffy Growth (K)	0.27	0.29	0.19	0.16
Trophic Level	3.82	0.55	3.80	0.54
Is Highly Migratory	0.09	0.29	0.05	0.21
Governance	1.06	0.81	1.42	0.64
Policy Start Year			1996	9
Policy Duration			12	8
Fisheries	3586		305	
Observations	120,332		14,697	

The resulting dataset, summarized in Table 1, has 120,332 observations from 3586 fisheries, 305 of which enter into IQ management during the study window.[17] Of those 3586 fisheries, 2399 (67%) experience collapse at some point during the sample, 980 of which spend fewer than 5 years of the sample collapsed. Collapse events last 4.6 years on average, with a median of 2 years and a mode of 1.[18]

Overall, rates of collapse are lower for IQ-managed fisheries, though less so in years during which IQ management is in place. The latter comparison does not indicate failure of IQ management: time trends in both collapse (2) and IQ introduction hint that simple correlation may be spurious. Among IQ-managed fisheries, catch rates are lower under the policy than just before, consistent with managers under the new policy imposing more responsible quotas which could help ward off collapse.

4.3. Estimation strategy

To illustrate potential benefits of the proposed causal forest method, I estimate the effect of IQ adoption on the probability of fishery collapse in two ways. First, I estimate standard parametric specifications, then I apply the causal forest methods described above. Both approaches share the same identification strategy, model of selection into treatment, and estimation target.

Estimating parametric variants before using causal forests offers three main benefits. First, estimating a model analogous to a primary specification used in Costello et al. (2008) provides suggestive evidence that any differences in results obtained from the new methods presented here are not likely attributable to the updated dataset. To this end, I estimate a linear probability model relating collapse to a constant and three regressors: whether a fishery is ever an IQ, a year effect, and how many years the fishery has been under IQ management. Second, I conduct falsification tests using those established methods to further probe the identification strategy which the causal forest method shares. Finally, examining results from alternate parametric specifications that encode different types of heterogeneity and dynamics highlights specification challenges in fully parametric approaches.

After estimating the parametric models, I apply the modified causal forest method to generate the primary results of the paper. The most general specification allows policy effects to vary with five dimensions: pre-treatment relative catch, pre-treatment relative catch trends, and Von Bertalanffy growth rate of the target species, policy duration, and policy start year. For comparison, I also present estimates in which the policy effect is allowed to vary in a subset of those dimensions. In each case, I estimate a causal forest with 20,000 trees, and each tree includes a random subset of 20% of fisheries, with sampling stratified by eventual IQ adoption.

To account for likely endogenous adoption of an IQ policy, the probability of transitioning to IQ management is estimated as a pooled logit model using the set of observations that are not already under IQ management in a given year. Explanatory variables include lagged catch (both absolute and relative), lagged catch trend (absolute), the maximum historical catch, year effects, and a linear time trend in the effect of the lagged relative catch. The logic behind such a model is that catch history is likely to influence adoption of a new management policy. More flexible methods could be used to estimate probabilities of policy transition, such as gradient boosting. I present logit results because the resulting balance is acceptable (see below) and for computational tractability.

During the course of estimation, after propensity scores are estimated, records are eliminated in which estimated propensity scores are numerically equivalent to zero or one in order to satisfy overlap requirements. In addition, trimming is used to eliminate records with probability of observed treatment sequence below 0.001.

[17] For models including species growth rates, data availability limits the sample to 89,523 observations from 2618 fisheries, with 250 IQ-managed fisheries. Many of the observations without available growth rates correspond to shellfish fisheries for which Von Bertalanffy growth is not appropriate.

[18] Autocorrelation in collapse across years is 0.42 for a lag of 1 year, dropping to 0.11 after 5 years.

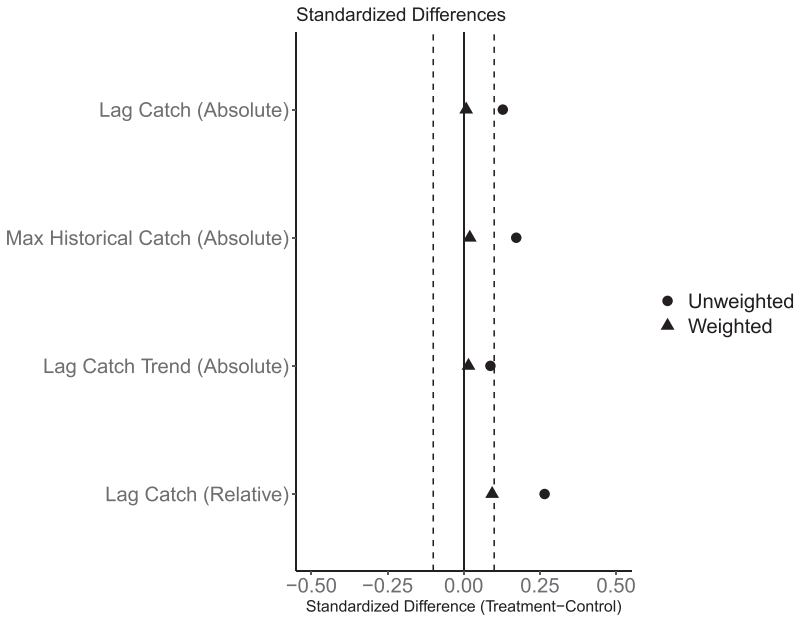

Fig. 3. Standardized differences of covariates across non-IQ and IQ observations before and after reweighting by stabilized inverse probability of transition from non-IQ to IQ management.

4.4. Assessing propensity score methods

The internal validity of the proposed approach relies on reweighting to render treatment as good as random. To assess balance of pre-policy covariates between fisheries selected into treatment in a given year and those not, I compute standardized differences in the means of those covariates. Fig. 3 presents those standardized differences in means of covariates before and after reweighting; all standardized differences are smaller in absolute value than 0.1 (Stuart et al., 2014; Austin and Stuart, 2015). Similarly, Fig. 4 presents balance plots for the propensity score and covariates used in the propensity score model before and after reweighting by the inverse probability of selection into an IQ program. The 25th, 50th, and 75th quantiles are all similar between treated and control observations after reweighting. Note that these assessments use weights based on the probability of transitioning to IQ management, whereas the forest procedure uses weights based on the probability that an observation experienced its entire treatment history. Assessing balance using the probability of selection in a given year allows a single balance assessment versus one per potential treatment exposure. Moreover, the maintained assumption of permanent treatment – satisfied in this dataset – limits concern over endogeneity to initial program adoption. As such, assessing balance once using reweighting by the probability of selection into treatment is sufficient.

In addition to these standard balance assessments, section 4.5.1 presents parametric falsification tests examining whether IQ adoption impacts outcomes which should not be affected under the assumptions outlined earlier.

4.5. Results

4.5.1. Standard parametric estimation

One of the primary results from Costello et al. (2008) suggests that the baseline probability of fishery collapse increases through time, but each additional year under IQ management can counteract and potentially completely offset that trend. Estimating an analogous model[19] on the updated dataset (restricted to $t \leq 2003$ for consistency with Costello et al. (2008)) yields encouragingly similar results. The marginal effects are presented in column 1 (unweighted) and column 2 (inverse probability of treatment weighted) of Table 2. While the probability of collapse increases around half a percent a year, an additional year of IQ management reduces the probability of collapse by around the same amount. These results are consistent with the main results presented in Costello et al. (2008), suggesting the datasets are sufficiently similar up to 2003 that substantive differences in results from the tree-based methods are likely attributable to the methods themselves.

Estimating variants of this basic model with alternate outcomes provides no evidence of identification concerns. Specifically, I estimate model variants with 1) the outcome variable in the prior year, 2) outcome variables randomized across fisheries within a year, and 3) outcome variables randomized across fisheries within a region (LME) in a year. None of these models should

[19] I use linear probability models, but logit specifications as in Costello et al. (2008) provide similar results.

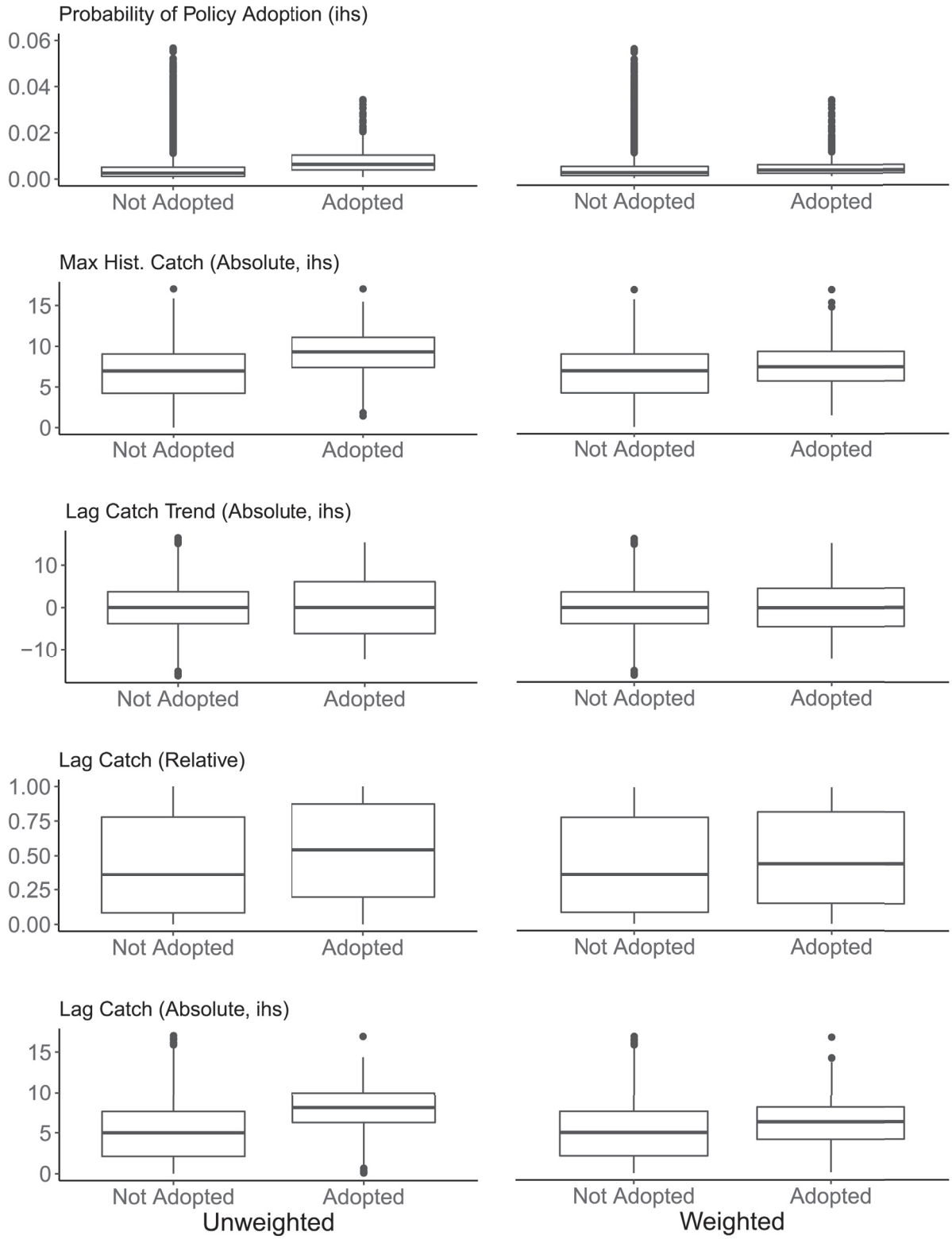

Fig. 4. Propensity score and covariate balance across treatment status before (left column) and after (right column) reweighting by stabilized inverse probability of transition from non-IQ to IQ management.

Table 2
Linear probability model estimation of collapse using (1) model in Costello et al. (2008) unweighted, (2) the same model weighted by inverse probability of observed treatment sequence, (3) interacting treatment with policy start year rather than duration, (4) interacting treatment with duration, policy start year, and pre-policy relative catch levels and trends, and (5) specification (4) plus allowing duration effects to depend upon policy start year, and pre-policy relative catch levels and trends.

	(1)	(2)	(3)	(4)	(5)
Ever IQ	−0.159***	−0.112***	−0.211***	−0.025	−0.062
	(0.020)	(0.023)	(0.015)	(0.040)	(0.032)
Year	0.006***	0.008***	0.005***	0.005***	0.005***
	(0.000)	(0.001)	(0.000)	(0.000)	(0.000)
Is IQ × Years in IQ	−0.005*	−0.008***		−0.002	0.009
	(0.002)	(0.002)		(0.002)	(0.006)
Is IQ × (Policy Start Year-1973)			0.006**	0.004	0.006*
			(0.002)	(0.002)	(0.003)
Is IQ × (Pre-Policy Relative Catch)				−0.274***	−0.219***
				(0.063)	(0.063)
Is IQ × (Pre-Policy Relative Catch Trend)				0.072	0.097
				(0.113)	(0.124)
Is IQ × Years in IQ × (Policy Start Year-1973)					−0.000
					(0.000)
Is IQ × Years in IQ × (Pre-Policy Relative Catch)					−0.013
					(0.007)
Is IQ × Years in IQ × (Pre-Policy Relative Catch Trend)					−0.001
					(0.012)

$^{***}p < 0.001, ^{**}p < 0.01, ^{*}p < 0.05$.

Table 3
Falsification tests.

	(1)	(2)	(3)
Ever IQ	0.053	0.000	0.021
	[−0.001; 0.107]	[−0.052; 0.055]	[−0.029; 0.084]
Year	−0.006[a]	0.001	−0.000
	[−0.011; −0.001]	[−0.002; 0.003]	[−0.004; 0.003]
Is IQ × Years in IQ	0.003	−0.000	0.000
	[−0.007; 0.013]	[−0.006; 0.005]	[−0.006; 0.006]

Notes: Examination of the influence of IQ duration on outcomes it should not effect: (1) outcome equal to relative catch in prior year, (2) outcomes randomized across fisheries within a year, and (3) outcomes randomized across fisheries in the same LME and same year.
[a] 0 outside the confidence interval.

display significant effects of treatment duration on the outcome of interest; the first is an assessment of reverse causality, while the second and third evaluate potential spillovers across fisheries. As indicated in Table 3, there is no significant evidence of policy duration affecting any of these variables.

One concern with the parametric estimation is that the way in which the policy effect varies may be misspecified. As one example, longer durations of policy exposure are only experienced in fisheries subjected to early implementations of IQs. If those early implementations were more effective, the apparent improvement in IQ impact with duration could be spurious. A model instead including an interaction between IQ management and the policy start year is consistent with this line of reasoning (Table 2 column (3)). More broadly, interacting the policy dummy with duration, the start year, and pre-policy catch levels and trends simultaneously indicates the relationship between duration and effectiveness may not be robust (Table 2 column (4)). Instead, pre-policy catch levels appear to play a larger role. Even when the effect of policy duration is itself allowed to be heterogeneous, more closely aligning with the flexibility afforded by the causal forest approach, duration still appears to play a limited role (Table 2 column (5)).

The changing estimates across the columns of Table 2 are a hallmark of misspecification issues, demanding a more structured approach to understanding variation in policy effects. Further, the handful of parametric specifications estimated do not even touch on potential nonlinearities in how the policy effect varies. Alternative models might interact the policy variable with one or more threshold-based indicator variables or polynomials. One view of this challenge is that the modeling task is actually to identify potentially low-dimensional heterogeneity in a higher-dimensional space defined by nonlinear transformations of a few regressors. Tools often associated with high-dimensional problems, including causal forests, thus may have great value, especially when detection of effects is challenged by the limited variation left over after the inclusion of fixed effects and numerous controls. With this motivation I turn to the causal forest results.

4.6. Causal forest methods

Applying the causal forest approach yields a nuanced picture of IQ effectiveness. I first allow partitioning based only on policy duration, yielding results that are broadly consistent with those from Costello et al. (2008): over the course of 30 years of IQ

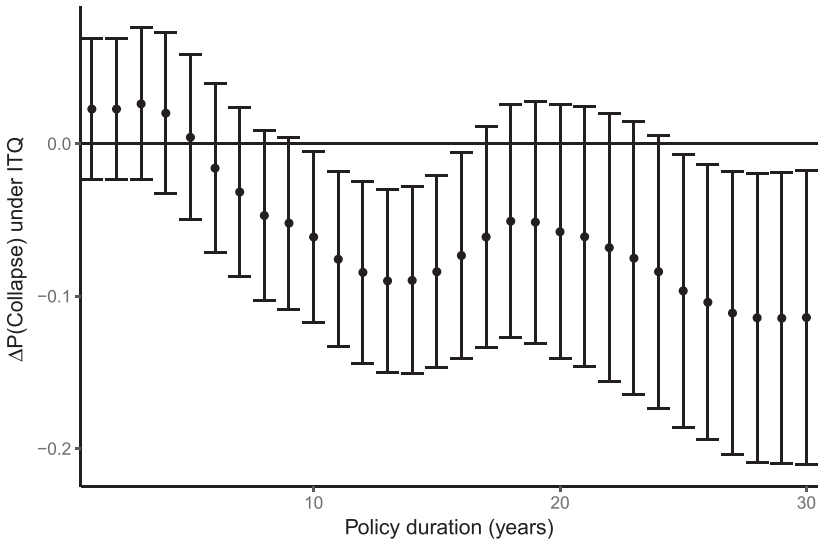

Fig. 5. Change in probability of collapse associated with IQs as a function of the number of years IQ management has been in effect. Points represent mean estimates and error bars coincide with 95% confidence intervals, with all quantities computed via infinitesimal jackknife adjusted for panel subsampling at the fishery level.

implementation, the probability of collapse declines *on average* by roughly a half percent per year (Fig. 5). However, even in this restricted setting, the method suggests there are substantial nonlinearities in how policy effects depend on the length of exposure. Expectations of immediate benefits may need to be tempered, and that reductions in the probability of collapse may not be steady. Note that while the confidence intervals for the many estimated policy effects overlap, due to correlations among the estimates, many differences between policy effects are, in fact, significant. For example, estimated reductions in collapse are significantly larger after 7–17 years or more than 22 years of IQ management than they are after one year.

Allowing policy effects to depend not only on duration, but also on start year and pre-policy characteristics reveals a richer picture, albeit one in which the role of duration is substantially muted. Because the estimated policy effect is a function of several variables, I summarize the results in a series of grouped partial dependency plots. Fig. 6 plots estimated policy effects against duration, depicting effect profiles for groups of fisheries that vary in some other dimension. For example, panel (A) depicts partial effects of policy duration at three different levels of pre-treatment relative catch (10%, 55%, and 80%). When catch is already low prior to IQ implementation, the new policy actually has little effect on the probability of collapse. The strongest benefits of the policy accrue when pre-treatment catch is at moderate or high levels, with small improvements in the probability of collapse through time.

Panel (B) of Fig. 6 offers additional evidence consistent with the claim that the effect of IQ adoption depends on the resource status prior to the policy change. Effects are weakest for fisheries with stable catch levels prior to adoption and stronger in fisheries with changing catch levels, especially those in which catch was declining prior to adoption. One plausible explanation for these differences is that stable pre-policy catch could reflect more active management (e.g. the use of fishery-wide quotas) prior to IQ adoption, which would dampen some of the potential gains from IQ implementation.

Panel (C) of Fig. 6 examines interactions between the two time dimensions, suggesting that IQ policies have had a less beneficial impact in more recent implementations. The estimated reduction in the probability of collapse shifts upward, flattens in time profile, and loses significance for implementations that began in more recent years. While this apparent lack of effectiveness in more recent years could be concerning, not all IQ programs have (exclusively) biological objectives. More recent adoptions may target economic goals rather than biological objectives; the latter may already have been achieved through other management strategies. Unfortunately, assessing impacts on profitability or other economic outcomes of interest is precluded by data availability at the scale of this analysis.

The relatively flat profiles of the policy effects in Fig. 6 also hint that the improvement of IQ effectiveness with policy duration may not be as pronounced as suggested by Fig. 5 or the study it approximates (Costello et al., 2008). Instead, point estimates for policy effects appear to vary more with pre-IQ catch levels and the year an IQ was introduced. In particular, the panel (C) of Fig. 6 suggests that for identical policy durations, earlier implementations are associated with a greater reduction in the probability of collapse, but that duration matters relatively little. Because longer policy durations must be observed in fisheries in which IQs began in earlier start years, these results suggest the apparent relationship between policy effects and duration in Costello et al. (2008) may actually reflect other interactions. Earlier IQ implementations with longer duration also tended to have higher and

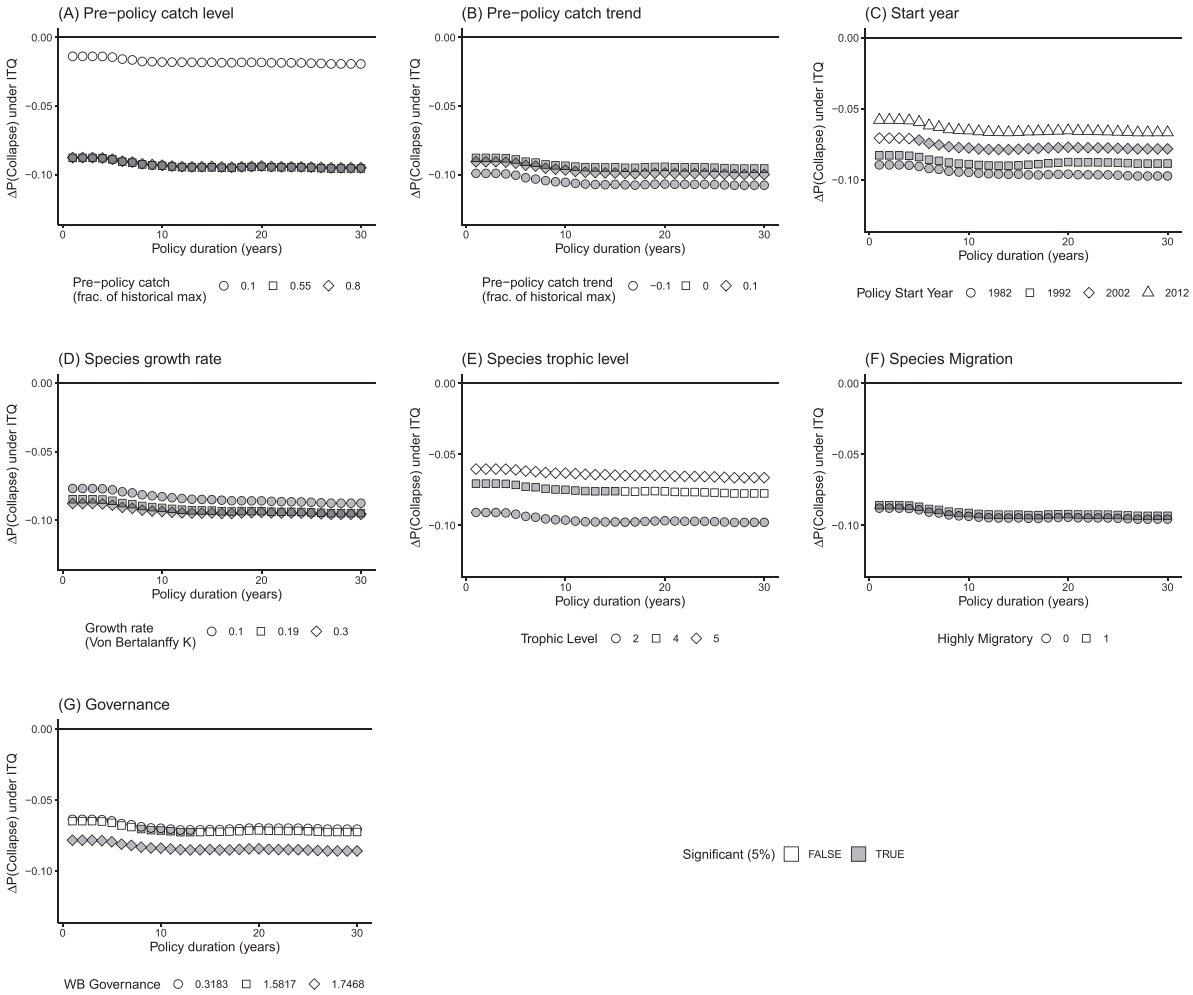

Fig. 6. Change in probability of collapse associated with IQs as a function of policy duration, with different groupings per panel. Filled (unfilled) shapes are significant (not significant) at the 5% level. Effects grouped by (A) pre-policy relative catch, (B) pre-policy relative catch trend, (C) start year, (D) target species growth rates, (E) trophic level (lower left), and (F) highly migratory status, as well as (G) World Bank governance score. When not used as a grouping variable for a panel, covariates take on the following values: start year is 1986, pre-policy catch relative catch is 0.55, pre-policy relative catch trend is 0, growth rate is 0.3, trophic level is 3.3, the species is not highly migratory, and governance is 1.85. For (C), only effects for treatments observed in the dataset are shown, e.g. the effect of more than 3 years of treatment for a fishery entering IQ management in 2012 cannot be directly estimated from the dataset, which ends in 2014.

less stable pre-policy catch levels, which would also yield the observed association between policy effects and duration.[20]

I next turn to species characteristics. Variation in policy effects with species growth rates follows sensible patterns (Fig. 6, panel (D)). The fastest growing species display reductions in the probability of collapse sooner, though effects for moderate growth species eventually reach comparable levels. That finding is in line with the results in panel (E) of Fig. 6, which indicates larger effects for lower trophic level species, which often have shorter generation times. If (higher) trophic level is a proxy for (higher) value, these findings indicate that higher gains from enforcing property rights under IQ do not necessarily translate to a greater reduction in collapse. One possible explanation could be that higher trophic level species depends on the management of prey species, so that transitioning a predator species to IQ management may have a smaller effect unless management of prey species also improves.

Finally, Fig. 6 also examines how policy effects depend upon governance, as proxied both by the migratory status of the target species and by a World Bank indicator. Intuition suggests that when species are highly migratory or when governance is poor, management is less likely to succeed in achieving biological objectives such as avoiding collapse. The patterns in panels (F) and

[20] The apparently minor role of duration is not an artifact of the specific covariate and start date profiles used in Fig. 6. Across all IQ fisheries in the dataset, duration influences the estimated policy effect by at most 1.2 percentage points.

(G) are consistent with this intuition as well as results in Isaksen and Richter (2019), though differences are minor at best.

Three key caveats on interpretation of these results are in order. The first applies to the causal forest method in general. The estimated treatment effects using causal forests have a causal interpretation if the key identifying assumption w-DCIA holds. Even if that assumption holds, however, *differences* in effects of a given policy exposure across groups (heterogeneity) or across different policy exposures (duration or start time) need not be caused by the covariate values that best predict those differences. Causal forests identify patterns of variation in treatment effects; those patterns can inform more targeted studies designed to identify causes of variation in treatment effects.

A second caveat is specific to interpretation of the treatment effect in this empirical application. In some fisheries, the introduction of individual quotas coincided with the first introduction of any kind of binding harvest quota, whether individual or fishery-wide. The estimated treatment effect may thus reflect many changes, including the introduction of any sort of quota, enforcement mechanisms and associated institutions, and the specific use of individually-assigned rights. Nevertheless, the preceding analysis highlights heterogeneity, dynamics, and time trends in how the probability of collapse under IQ management differs from that without IQ management.

Third, I note that while the focus of these results and the paper is on how time (duration and policy start year) moderates policy effects, cross-sectional characteristics may also interact in their influence on the effectiveness of IQ programs. The causal forest approach can equally be used to examine those interactions; I provide an example in the online appendix (Section S3).

4.7. Alternate outcomes

To address concerns raised earlier about the definition of collapse used here and in earlier work, I also estimate causal forest models with the inverse hyperbolic sine of catch as an outcome rather than collapse. All other aspects of estimation remain unchanged. Fig. 7 suggests that IQs increase catch, with greater effects for stocks with higher relative catch prior to IQ adoption (top left), for earlier implementations (bottom left), and for faster growing species (bottom right). Differences across fisheries based on pre-policy catch trends are less pronounced (upper right). Overall, these findings are qualitatively consistent with the reductions in the probability of collapse reported earlier. Policy effectiveness continues to exhibit only a limited relationship with policy duration.

5. Conclusion

This paper has shown how to adapt a recently proposed suite of methods – causal trees (Athey and Imbens, 2016) and causal forests (Wager and Athey, 2018) – to simultaneously examine cross-sectional heterogeneity, dynamics, and changes in policy effectiveness through time. The estimation approach uses a modified random forest procedure to identify subpopulations in which the treatment effect can be approximated by a constant, and makes use of inverse probability of treatment weighting to construct estimates in each subsample. One key benefit of this method is that it requires weak assumptions about the way in which policy effects vary. The researcher need only specify which variables might influence policy effects, but not the functional form of that influence. This allows data to potentially reveal a richer pattern of heterogeneity and time dependence than what may be hypothesized. In addition, by partitioning the data to target differences in the estimated policy effect function, the approach may improve policy effect detection in observational settings when residual variation is limited after removing the influence of confounders.

Applying these methods to examine the effect of individual quota programs on fishery collapse reveals heterogeneity across fisheries and variation through time. Individual quota programs successfully reduce the probability of fishery collapse for some fisheries, while yielding little benefit for others. Moreover, the apparent strengthening of impacts with policy duration identified in prior work may be driven as much or more by policy start dates or pre-policy fishery characteristics. These findings suggest that application of causal forests to other environmental policy questions could reveal heterogeneity or dynamics that have been previously ignored or assumed away.

More generally, modified causal forests may prove useful in many empirical studies of both environmental policies and shocks. Environmental policies are likely to impact people, firms, or jurisdictions in different ways, and those effects may vary depending on when a policy first took effect and for how long it has been in place. Learning more about heterogeneity, dynamics, and evolution of policy effects has many potential benefits. At a minimum, we can set expectations as to when policy effects might start to arise for a new policy and who stands to benefit the most from it. Knowledge of heterogeneity in policy effects might also help better target a particular type of policy to settings in which it is likely to have the most beneficial effect. Methods permitting more effective detection of nuanced differences in policy impacts, including causal forests, can help us in those efforts.

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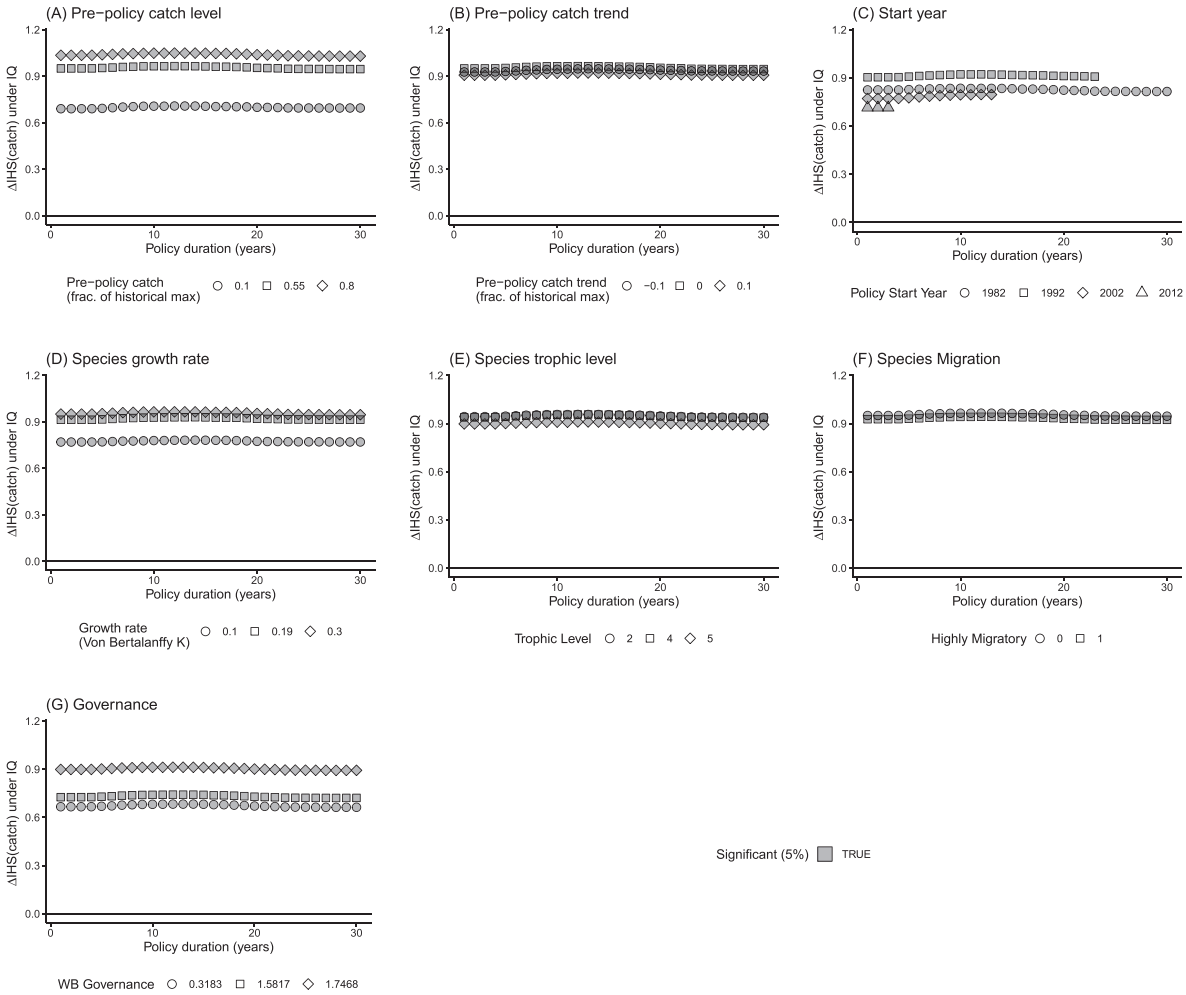

Fig. 7. Change in the inverse hyperbolic sine of catch associated with IQs as a function of policy duration. Other than the change in outcome variable, the plot is analogous to Fig. 6. Filled (unfilled) shapes are significant (not significant) at the 5% level. Upper left panel: effects grouped by pre-policy relative catch, for a policy start year of 1982 and a sample mean of pre-policy relative catch trends. Upper right panel: effects grouped by pre-policy relative catch trend, for a policy start year of 1982 and pre-policy relative catch of 0.5. Lower left panel: effects grouped by start year, for pre-policy relative catch of 0.5 and the sample mean of pre-policy relative catch trends. Note only effects for treatments observed in the dataset are shown, e.g. the effect of more than 3 years of treatment for a fishery entering IQ management in 2012 cannot be directly estimated from the dataset, which ends in 2014. Lower right panel: effects grouped by target species growth rates, for pre-policy relative catch of 0.5, sample mean of pre-policy relative catch trends, and a policy start year of 1982.

Declaration of competing interest

I have no conflicts of interest to disclose for this paper.

A. Appendix

A.1. Within-subsample estimator

For given values of d and s and a subpopulation defined by $X_{is-1} \in \mathcal{X}$, the within-subsample estimator is a standard IPTW estimator:

$$\widehat{\tau}(\mathcal{X}, d, s) = \sum_{i,t:\, \substack{t=s+d-1,\\ \mathbf{X_{is-1}} \in \mathcal{X}}} w_{ist}(d) y_{it} - \sum_{i,t:\, \substack{t=s+d-1,\\ \mathbf{X_{is-1}} \in \mathcal{X}}} w_{ist}(0) y_{it}, \tag{13}$$

$$w_{ist}(\widetilde{d}) = \frac{\frac{1(D_{it}=\widetilde{d})}{\widehat{p}_i(d,s,t)}}{\sum_j \frac{1(D_{jt}=\widetilde{d})}{\widehat{p}_j(\widetilde{d},s,t)}} \text{for} \widetilde{d} \in \{0, d\}. \tag{14}$$

In short, $\widehat{\tau}(\mathcal{X}, d, s)$ is a weighted difference in mean outcomes between entities experiencing exposure (s, d) and those experiencing $(s, 0)$. Weights are normalized per group.

Dataset construction

The list of fisheries subjected to IQ management was based on the database of catch share programs provided by the Environmental Defense Fund. Since that database provides a single start year for an entire management program, which may involve multiple species, start dates for each species in each LME were subsequently compiled from published government sources or academic research papers. In some cases, a single species may be subjected to several forms of management in a single LME. Some LMEs span national jurisdictions, and even within a jurisdiction and LME, different groups of vessels (e.g. those using different fishing gear) may be subjected to different management. The start date used for IQs in such cases was the earliest year in which the species was subjected to IQ management in that LME. A more refined definition of a fishery could be used to avoid this issue, but defining a fishery as an LME × species pair allows for a clean comparison with Costello et al. (2008).

As explained in the main text, species name and LME were used to match management information with catch records. Reported landings from the Sea Around Us project were downloaded per LME. Records from the management database without an exact (species × LME) match in the catch records were manually reviewed. Where possible, exact matches were supplemented with manual matches, correcting differences in spelling or the use of non-standard species synonyms in one source or the other. Catch records were matched to trait data from FishBase based on species names alone. If no exact match was found, several additional steps were used to increase coverage. First, the species name from the catch data was matched against genus names in FishBase. In some cases, catch data was actually reported at the genus level, so that the 'species' name actually represented a genus. Genus traits were computed as means and modes of species-level traits for species in that genus. Supplementary steps included attempting matches based on synonyms for species names used in the catch data, as well as manually mapping taxonomic family names used in catch data to a representative genus for matching to the genus-level traits just described.

Appendix B. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jeem.2020.102337.

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